

Zhen Chen

Adobe Research, Seattle, WA

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Research Interests

Physics-based simulation, shell modeling, generative models, and robust geometry processing.

Employment

Adobe	Seattle, WA
Research Engineer II	2026 – Present
Research Scientist/Engineer	2024 – 2026

Education

The University of Texas at Austin	Austin, TX
Ph.D. in Computer Science	2018 – 2024
Advisor: Prof. Etienne Vouga	
University of Science and Technology of China	Hefei, China
B.S. in Information and Computing Science	2014 – 2018
Department of Information and Computing Science, School of Mathematics	

Publications

- [1] **Zhen Chen**, Daniele Panozzo, Jeremie Dumas. **Half-Space Power Diagrams and Discrete Surface Offsets**. *IEEE Transactions on Visualization and Computer Graphics*, 2019.
- [2] **Zhen Chen**, Hsiao-yu Chen, Danny M. Kaufman, Mélina Skouras, Etienne Vouga. **Fine Wrinkling on Coarsely-Meshed Thin Shells**. *ACM Transactions on Graphics (SIGGRAPH)*, 2021.
- [3] Yan Zheng, Lemeng Wu, Xingchao Liu, **Zhen Chen**, Qiang Liu, Qixing Huang. **Neural Volumetric Mesh Generator**. *NeurIPS 2022 Workshop*, 2022.
- [4] **Zhen Chen**, Danny M. Kaufman, Mélina Skouras, Etienne Vouga. **Complex Wrinkle Field Evolution**. *ACM Transactions on Graphics (SIGGRAPH)*, 2023.
- [5] **Zhen Chen**, Zherong Pan, Kui Wu, Etienne Vouga, Xifeng Gao. **Robust Low-Poly Meshing for General 3D Models**. *ACM Transactions on Graphics (SIGGRAPH)*, 2023.
- [6] Josh Vekhter, **Zhen Chen**, Etienne Vouga. **MiNT: Discretely Integrable Moments for Symmetric Frame Fields**. *Symposium on Geometry Processing*, 2025.
- [7] Gonzalo Gomez-Nogales, **Zhen Chen**, Rosalie Martin, Elena Garces, Danny M. Kaufman. **Unlocking Thickness Modeling for Codimensional Contact Simulation**. *arXiv preprint*, 2025.
- [8] Ming Li, Hui Shan, Kai Zheng, Chentao Shen, Siyu Liu, Yanwei Fu, **Zhen Chen**, Xiangru Huang. **ReWeaver: Towards Simulation-Ready and Topology-Accurate Garment Reconstruction**. *The IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026.

Patents

- [1] **ISO-surface mesh generation for three-dimensional models.** *US 12,505,621*
[2] **Low-poly mesh generation for three-dimensional models.** *US 12,333,653*

Industry Experience

Research Scientist/Engineer, Adobe

Seattle, WA

Manager: Danny M. Kaufman

May 2024 – Present

- Main contributor to dynamic physics presets for Adobe Express.
- Main developer of the internal simulator engine.
- Contributor to *Project Generative Physics*.

Research & Development Intern, Tencent America

Bellevue, WA

Mentor: Xifeng Gao

Summers 2022, 2023

- Developed robust algorithms for convex decomposition of 3D meshes.
- Designed remeshing methods for sharp feature preservation and watertightness.

Research Intern, Adobe

Remote / Austin, TX

Mentor: Danny M. Kaufman

Summer 2021

- Designed a time integrator for IPC models to balance dissipation and dispersion.

Academic Experience

Graduate Research Assistant, The University of Texas at Austin

Austin, TX

Supervisor: Etienne Vouga

2019 – 2024

- Worked on thin shell simulation, geometry processing, volumetric model generation, and robust remeshing.

Student Intern, Geometric Computing Lab, NYU

New York, NY

Host: Daniele Panozzo

Summer 2017

- Worked on robust computation of offset surfaces for 3D meshes using half-space power diagrams.

Talks

Complex Wrinkle Field Evolution

SIGGRAPH 2023

Robust Low-Poly Meshing for General 3D Models

SIGGRAPH 2023

Fine Wrinkling on Coarsely-Meshed Thin Shells

SIGGRAPH 2022

Half-Space Power Diagrams and Discrete Surface Offsets

SGP 2020

Professional Service

SIGGRAPH

2023 – 2026

SIGGRAPH Asia

2024, 2025

Eurographics

2024

Computer Graphics Forum

2022

IEEE TVCG

2023, 2025

NeurIPS

2024

SCA	2024 (IPC Member)
Pacific Graphics	2025
ICLR	2025, 2026
ICML	2025
AISTATS	2025
CVPR	2026

Teaching Experience

Teaching Assistant, Department of Computer Science, UT Austin	
CS 303E: Elements of Computers and Programming	Fall 2018, Fall 2021
Teaching Assistant, Department of Mathematics, USTC	
Complex Analysis	Fall 2017
Mathematical Analysis	Spring 2017

Honors and Awards

Baosteel Scholarship (Top 2%)	2017
National Scholarship (Top 1% nationwide)	2016
Outstanding Freshman Scholarship (Top 1%)	2014

Skills

- Programming:** C/C++, Python, Matlab
- Software:** Houdini, Adobe Premiere
- Languages:** Chinese (native), English (fluent)